

Relative Exponential Fixed Parts and Functional Periods: An Exponential Nori–Ayoub Theorem

Christopher D. Long

Antoine-Auguste Le Blanc*

August 2026

Abstract

We construct a relative category of exponential Nori motives over a smooth algebraic base and prove an exponential analogue of the Nori–Ayoub functional period theorem. For a lisse relative exponential motive, the maximal trivial subconnection of its exponential Gauss–Manin realization is induced by a maximal constant exponential submotive. Over $\overline{\mathbb{Q}}$, for every finitely generated Tannakian subcategory of the generic relative Kontsevich–Zagier exponential envelope and every $\lambda \in \overline{\mathbb{Q}}^\times$, we obtain an exact sequence

$$1 \longrightarrow G_{\text{diff}} \longrightarrow G_{\text{mot}} \longrightarrow G_{\text{const}} \longrightarrow 1.$$

At $\lambda = 1$, the generic relative formal exponential-period map is injective after imposing the algebraic relations among constant exponential periods.

The proof combines Sabbah’s relative Fourier–nearby-cycle comparison with the arithmetic theory of 1-motives of Huber–Wüstholz to control differential characters. Numerical relations arising only after specialization, and an elementary presentation of all motivic period relations, are separate questions.

Contents

1	Introduction	2
1.1	The functional problem	2
1.2	Main results	3
1.3	Organization	4
2	Background and conventions	4
2.1	Perverse Nori motives	4
2.2	Exponentiation	5
2.3	Rapid decay and exponential periods	5
3	Consequences of Virk’s realization theory	6
4	The relative exponentiated Nori heart	6
5	The invariant fixed-part theorem	8

*Le Blanc cryptographic author identifier: b604812530b3535b23d6dbb0f5abe32d1ed1a9de35fb4f7d04d62cff08ea19c8

6	Differential characters and Fourier geometry	9
6.1	Rank-one lifting	9
6.2	A split Fourier-geometric criterion	10
6.3	Nonzero Fourier fibers and nearby cycles	11
6.4	Sabbah’s conventions and the relative comparison	12
6.5	The arithmetic exponential kernel	15
6.6	Generic saturation for the Kontsevich–Zagier category	16
7	Twisted fixed parts and Galois exactness	17
7.1	Semi-invariant spaces	17
7.2	Normality	18
8	The relative functional period theorem	18
8.1	Comparison torsors and the constant-period base	18
8.2	Comparison with Picard–Vessiot theory	19
9	Relation with Kontsevich–Zagier exponential periods	20
9.1	Arithmetic specialization	20
10	Applications and examples	21
10.1	Polynomial exponential integrals	21
10.2	Cyclotomic Gamma packets and the Rohrlich–Lang boundary	21
10.3	Nilsson– E packets	24
11	Concluding perspective	24

1 Introduction

1.1 The functional problem

For the arithmetic results we work over

$$k = \overline{\mathbb{Q}},$$

and let S be a smooth connected quasi-projective k -variety. The exponentiated category and the fixed-part theorem are defined over any algebraically closed subfield $k \subset \mathbb{C}$. We use the k -linear scalar extension of the rational Nori categories.

A relative exponential period is represented schematically by a family

$$s \mapsto \int_{\gamma_s} e^{-f_s} \omega_s,$$

where the de Rham class is computed with the twisted differential $d - df_s \wedge$ and the Betti class is a rapid-decay cycle. Exponential periods admit equivalent naive, cohomological, and Nori-motivic descriptions [9, 10, 11]. In families, the resulting functions are solutions of exponential Gauss–Manin systems and admit rapid-decay integral representations [15, 14].

For ordinary periods, Nori and Ayoub prove a functional form of the Kontsevich–Zagier period conjecture: the relative formal period algebra injects into the corresponding analytic function field. Equivalently, the relative motivic comparison torsor is governed by algebraic monodromy [22, 3, 4, 25, 8]. The aim of this paper is to establish the analogous statement for exponential motives.

Functional and numerical relations must be distinguished. The functional problem concerns identities valid on a dense open of the parameter space. The numerical problem asks whether additional relations appear after specialization to an algebraic point. The latter requires separate arithmetic input and is not addressed here.

1.2 Main results

Let $\mathrm{MN}(X)$ be the abelian category of perverse Nori motives on a quasi-projective k -variety X [16, 34, 35]. For

$$\pi_X : X \times \mathbb{A}^1 \longrightarrow X,$$

define

$$\mathrm{EM}_N(X) := \mathrm{MN}(X \times \mathbb{A}^1) / \pi_X^*[1] \mathrm{MN}(X). \quad (1.1)$$

Theorem 1.1 (Exponentiated perverse-Nori heart). *The essential image of*

$$\pi_X^*[1] : \mathrm{MN}(X) \longrightarrow \mathrm{MN}(X \times \mathbb{A}^1)$$

is a Serre subcategory. The quotient in (1.1) is the heart of the exponentiated derived coefficient system associated with $D^b(\mathrm{MN}(-))$. It carries the six operations and the additive-convolution tensor product. The Nori-to-Hodge realization descends to an exact faithful functor

$$R_H^{\mathrm{exp}} : \mathrm{EM}_N(X) \longrightarrow \mathrm{EM}_H(X)$$

compatible with these operations and with every nonzero exponential realization.

Fix $\lambda \in \mathbb{C}^\times$. Let S have dimension d , let $a : S \rightarrow \mathrm{Spec} k$ be the structure morphism, and let $N \in \mathrm{EM}_N(S)$ be λ -lisse. Set

$$N^{\mathrm{fix}} := \mathrm{im} \left(a^* {}^p H^{-d}(a_* N)[d] \longrightarrow N \right). \quad (1.2)$$

Theorem 1.2 (Invariant exponential fixed part). *The λ -realization of N^{fix} is the maximal trivial differential submodule of $\rho_\lambda(N)$. The object N^{fix} is the maximal constant subobject of N .*

Let $\eta = \mathrm{Spec} k(S)$ and let $\mathcal{T}_{\mathrm{KZ}}(\eta)$ denote the generic rigid Tannakian envelope generated by elementary relative Kontsevich–Zagier exponential data.

Theorem 1.3 (Relative exponential Galois exactness). *Let $\lambda \in \overline{\mathbb{Q}}^\times$ and let $\mathcal{T}$ be a finitely generated rigid Tannakian subcategory of $\mathcal{T}_{\mathrm{KZ}}(\eta)$. Then the relative differential Galois group is normal in the motivic Galois group, and*

$$1 \longrightarrow G_{\mathrm{diff}} \longrightarrow G_{\mathrm{mot}} \longrightarrow G_{\mathrm{const}} \longrightarrow 1$$

is exact.

Theorem 1.4 (Relative exponential Nori–Ayoub theorem). *For the canonical realization $\lambda = 1$, after the algebraic relations among constant exponential periods have been imposed, the generic relative formal exponential-period map of the filtered Kontsevich–Zagier Tannakian envelope is injective. The same statement holds for every $\lambda \in \overline{\mathbb{Q}}^\times$.*

The fixed-part theorem follows from the lowest algebraic direct image. The proof of Theorem 1.3 requires control of differential characters. Sabbah’s relative Fourier–nearby-cycle comparison places their Betti local systems in the ordinary geometric-origin category; Lam–Litt then gives finite monodromy, and Huber–Wüstholz’s theory of 1-motives produces the required exponential-motivic lifts. The resulting semi-invariant subobjects yield normality by André’s criterion, and the comparison torsor is identified with the Picard–Vessiot torsor.

1.3 Organization

Section 2 reviews the coefficient systems and realizations. Section 3 records two consequences of Virk’s realization theory. The relative exponential Nori heart is constructed in Section 4, and the fixed-part theorem is proved in Section 5. Differential characters are treated in Section 6. The Galois exact sequence and functional period theorem appear in Sections 7 and 8, and Section 9 states the Kontsevich–Zagier consequence. The examples in Section 10 include a complete functional presentation for finite cyclotomic Gamma packets and isolate the classical Rohrlich–Lang problem in the constant Gamma values.

2 Background and conventions

2.1 Perverse Nori motives

Let $\mathrm{MN}_{\mathbb{Q}}(X)$ be the rational category of perverse Nori motives constructed by Ivorra–Morel. Throughout the paper we use its scalar extension

$$\mathrm{MN}(X) := \mathrm{MN}_{\mathbb{Q}}(X) \otimes_{\mathbb{Q}} k,$$

and similarly scalar-extend the Hodge and exponential realization categories; we suppress the coefficient field from the notation. All functors used below are $\mathbb{Q}$ -linear and extend k -linearly; on lisse Tannakian subcategories this is the usual scalar extension [18]. We take k algebraically closed so that finite-order rank-one characters and their idempotents are defined over k . The Betti realization becomes a faithful exact functor

$$\mathrm{rat}_X : \mathrm{MN}(X) \longrightarrow \mathrm{Perv}(X^{\mathrm{an}}, k).$$

Its bounded derived categories carry the four operations and nearby and vanishing cycles [16]. Terenzi constructs a canonical closed symmetric monoidal structure and completes the six-functor formalism [34]. Tubach proves that the corresponding derived categories are the derived categories of their constructible hearts and constructs operation-compatible Nori and Hodge realizations [35]. We use the following consequences.

- (N1) rat_X is faithful and exact, and its derived extension is conservative.
- (N2) Smooth pullback with the conventional relative-dimension shift is t -exact.
- (N3) The six operations, tensor product, duality, and perverse cohomology commute with Betti realization.
- (N4) There is a faithful exact Hodge realization

$$R_{H,X} : \mathrm{MN}(X) \longrightarrow \mathrm{MHM}(X)$$

whose derived extension commutes with the six operations.

The categorical construction and fixed-part theorem descend from the rational category by exact scalar extension. The arithmetic exponential-kernel theorem and the functional period theorem are stated over $\overline{\mathbb{Q}}$.

2.2 Exponentiation

Let $\mathcal{C}$ be a six-functor coefficient system. Gallauer–Pepin Lehalleur associate with it an exponentiated coefficient system $\mathcal{C}_{\text{exp}}$ whose objects over X may be represented on $X \times \mathbb{G}_a$, modulo objects pulled back from X . The tensor product is additive convolution in the potential coordinate [12]. In Virk’s Hodge/ D -module model one sets

$$E(X) = D(X \times \mathbb{G}_a) / \pi_X^* D(X),$$

and the quotient inherits the six operations and convolution [36, Sections 3–6].

For each $\lambda \in \mathbb{C}^\times$, Virk defines a realization

$$\text{real}_\lambda : E(X) \longrightarrow D_h^b(X)$$

by twisting with the exponential kernel $e^{-\lambda t}$ and integrating in the t -direction. He proves that it commutes with the four operations and duality, is t -exact, and is faithful on the exponential heart [36, Theorems 6.13, 7.5, and 7.6].

2.3 Rapid decay and exponential periods

For a smooth variety X , a regular function f , and an algebraic differential form ω , the twisted de Rham complex uses $d - df \wedge$. Its topological dual is rapid-decay homology. Hien proves the perfect period pairing for flat algebraic connections, and Hien–Roucairol identify solutions of exponential Gauss–Manin systems with rapid-decay integrals [14, 15]. Fresan–Jossen’s exponential Nori category packages the de Rham, rapid-decay, and perverse realizations [11]. Commelin–Habegger–Huber and Commelin–Huber compare the naive, cohomological, and motivic notions of exponential period [9, 10].

Proposition 2.1 (Absolute rapid-decay fiber functor). *For the absolute exponential Nori envelope over k , the perverse realization lands in Katz’s convolution category $\text{Perv}_0(\mathbb{A}^1)$. The nearby-fiber functor at infinity*

$$\Psi_\infty : \text{Perv}_0(\mathbb{A}^1) \longrightarrow \text{Vec}_k$$

is exact, faithful, and symmetric monoidal. On an elementary exponential pair, its value is canonically identified with rapid-decay Betti cohomology, compatibly with the twisted de Rham comparison.

Proof. This combines Katz’s convolution theorem with the perverse realization of Fresán–Jossen and its comparison with rapid-decay cohomology; see [21, 11] and [33, Corollary 2.17 and Theorem 3.6]. $\square$

For a geometric point $s \in S(\mathbb{C})$ and a lisse object $N \in \text{EM}_N(S)$, we use the rapid-decay Betti fiber

$$\omega_{B,s}(N) := \Psi_\infty(\text{rat}^{\text{exp}}(i_s^* N)),$$

where $i_s^* N$ is the absolute exponential Nori motive at s and Ψ_∞ is the nearby-fiber-at-infinity fiber functor on the perverse realization. For elementary exponential pairs this is the usual rapid-decay homology, and it is exact and monoidal on the lisse Tannakian subcategory [11, 14, 33]. The de Rham/rapid-decay comparison therefore supplies the comparison point used in Section 8.

3 Consequences of Virk's realization theory

The following consequences of Virk's realization theory separate the categorical and differential levels.

Proposition 3.1 (Derived conservativity and reflection of the t -structure). *Let $K \in E(X)$ and $\lambda \in \mathbb{C}^\times$. Then*

$$K \in E(X)^{\leq 0} \iff \mathrm{real}_\lambda(K) \in D_h^b(X)^{\leq 0},$$

and similarly for ≥ 0 . Thus real_λ is conservative on $E(X)$.

Proof. Virk proves t -exactness and faithfulness on the heart [36, Theorems 7.5–7.6]. Therefore

$${}^p H^i \mathrm{real}_\lambda(K) \simeq \mathrm{real}_\lambda({}^p H^i K).$$

If the left side vanishes, faithfulness on the heart gives ${}^p H^i K = 0$. Applying this degree by degree proves the assertions. $\square$

Proposition 3.2 (Exactness of the exponential Fourier transform). *Let $V \rightarrow S$ be a vector bundle. Virk's exponential Fourier transform*

$$\mathrm{FT}_V : E(V) \longrightarrow E(V^\vee)$$

is t -exact and induces an exact equivalence of exponential hearts.

Proof. Virk proves Fourier inversion and compatibility of FT_V with the classical D -module Fourier transform under real_1 [36, Theorems 5.3 and 6.12]. With Virk's shifts, the classical Fourier transform is t -exact. Hence the realization of $\mathrm{FT}_V(K)$ lies in the heart whenever K does. By Proposition 3.1, $\mathrm{FT}_V(K)$ lies in the exponential heart. Fourier inversion proves that the induced exact functor on hearts is an equivalence. $\square$

4 The relative exponentiated Nori heart

Fix a quasi-projective k -variety X and write

$$\pi = \pi_X : X \times \mathbb{A}^1 \longrightarrow X.$$

Since π is smooth of relative dimension one, $\pi^*[1]$ is t -exact on perverse Nori motives.

Lemma 4.1 (Reflection of constant objects). *Let $A \in \mathrm{MN}(X \times \mathbb{A}^1)$. If*

$$\mathrm{rat}(A) \simeq \pi^* L[1]$$

for some $L \in \mathrm{Perv}(X^{\mathrm{an}}, k)$, then there exists $B \in \mathrm{MN}(X)$ such that $A \simeq \pi^ B[1]$.*

Proof. Set

$$B := {}^p H^{-1}(\pi_* A).$$

Compatibility with Betti realization and $\mathbb{A}^1$ -homotopy invariance give

$$\mathrm{rat}(B) \simeq L.$$

The adjunction counit yields a morphism

$$\pi^* B[1] \longrightarrow A.$$

Its Betti realization is the canonical isomorphism $\pi^* L[1] \rightarrow \pi^* L[1]$. Since the derived Betti realization is conservative, the counit is an isomorphism. $\square$

Proposition 4.2 (Serre property). *The essential image of*

$$\pi^*[1] : \text{MN}(X) \longrightarrow \text{MN}(X \times \mathbb{A}^1)$$

is closed under subobjects, quotients, and extensions.

Proof. For perverse sheaves, the essential image of smooth pullback along a surjective smooth morphism with geometrically connected fibers is Serre [7, Corollaire 4.2.6.2]; Virk applies this to π in [36, Proposition 7.2]. Let A be a subobject, quotient, or extension of objects in the Nori essential image. Exactness of Betti realization places $\text{rat}(A)$ in the corresponding Serre subcategory of perverse sheaves. The conclusion follows from Lemma 4.1. $\square$

Lemma 4.3 (Derived constant subcategory). *Let*

$$\mathcal{C}_X := \pi^*[1]\text{MN}(X) \subset \text{MN}(X \times \mathbb{A}^1).$$

The exact functor $\pi^[1]$ is fully faithful, and its derived functor identifies $D^b(\text{MN}(X))$ with the full triangulated subcategory of $D^b(\text{MN}(X \times \mathbb{A}^1))$ consisting of complexes whose perverse cohomology objects lie in $\mathcal{C}_X$. This subcategory is stable under perverse truncations.*

Proof. Full faithfulness follows from adjunction and $\mathbb{A}^1$ -homotopy invariance. By Proposition 4.2, $\mathcal{C}_X$ is a Serre subcategory. Tubach’s derived-category theorem identifies the constructible derived category with the bounded derived category of its perverse-Nori heart. Standard d’évisage for a Serre subcategory then identifies $D^b(\mathcal{C}_X)$ with the full subcategory of complexes whose perverse cohomology lies in $\mathcal{C}_X$; see [7, Section 1.4]. The subcategory is therefore stable under perverse truncations. Finally, $\pi^*[1] : \text{MN}(X) \rightarrow \mathcal{C}_X$ is an equivalence, hence so are the bounded derived categories. $\square$

Definition 4.4 (Relative exponential Nori heart). Define

$$\text{EM}_N(X) := \text{MN}(X \times \mathbb{A}^1) / \pi^*[1]\text{MN}(X).$$

Theorem 4.5 (Exponentiated perverse-Nori heart). *The quotient in Definition 4.4 is the heart of*

$$D^b(\text{MN}(X \times \mathbb{A}^1)) / \pi^* D^b(\text{MN}(X)).$$

As X varies, these derived categories form the exponentiation of the perverse-Nori coefficient system. They carry the six operations and the additive-convolution tensor product. The Hodge realization descends to an exact faithful operation-compatible functor

$$R_H^{\text{exp}} : \text{EM}_N(X) \longrightarrow \text{EM}_H(X).$$

Proof. By Lemma 4.3, the thick subcategory being quotiented is stable under perverse truncations. The localization theorem for t -structures therefore endows the Verdier quotient with a bounded t -structure whose heart is the Serre quotient in Definition 4.4; compare [7, Section 1.4] and Virk’s proof in [36, Section 7]. Terenzi and Tubach provide a closed symmetric monoidal six-functor coefficient system on $D^b(\text{MN}(-))$ [34, 35]. Gallauer–Pepin Lehalleur’s exponentiation theorem therefore supplies the six operations and convolution on the quotient [12].

The Hodge realization commutes with pullback and carries $\pi^*[1]\text{MN}(X)$ into $\pi^*[1]\text{MHM}(X)$, so it descends to the quotient. It is exact because the source and target quotient functors are exact. To prove faithfulness, let $f : A \rightarrow B$ have zero image in the Hodge quotient. Then $R_H(\text{im } f)$

belongs to the Hodge constant Serre subcategory. Its underlying perverse sheaf is pulled back from X , so Lemma 4.1 gives $\mathrm{im} f \in \pi^*[1]\mathrm{MN}(X)$. Hence f was already zero in the Nori quotient. Compatibility with the operations is inherited from Tubach's realization and the functoriality of exponentiation. $\square$

For $\lambda \in \mathbb{C}^\times$ put

$$\rho_\lambda := \mathrm{real}_\lambda \circ R_H^{\mathrm{exp}}.$$

Definition 4.6 (Lisse and admissible Tannakian subcategories). An object $N \in \mathrm{EM}_N(X)$ is λ -lisse if $\rho_\lambda(N)$ is a vector bundle with integrable connection. The full λ -lisse subcategory is a Serre tensor subcategory of $\mathrm{EM}_N(X)$, because ρ_λ is exact, faithful, and tensor-compatible and algebraic connections form a Serre tensor subcategory of holonomic $\mathcal{D}$ -modules.

A λ -admissible Tannakian subcategory is a strictly full rigid tensor subcategory of the λ -lisse heart which is closed under subquotients. All Tannakian arguments below take place in such a subcategory. When the scale is fixed, we suppress λ .

Remark 4.7 (Absolute exponential motives). For $X = \mathrm{Spec} k$, recollement identifies the quotient with the Nori envelope generated by perverse objects on $\mathbb{A}^1$ with vanishing global cohomology, the model used by Fresán–Jossen for absolute exponential motives [11].

5 The invariant fixed-part theorem

Let S be smooth, connected, and of dimension d , let $a : S \rightarrow \mathrm{Spec} k$ be the structure morphism, and fix a scale $\lambda \in \mathbb{C}^\times$. Throughout this section $N \in \mathrm{EM}_N(S)$ is λ -lisse.

Definition 5.1 (Constant subcategory). The constant exponential subcategory of $\mathrm{EM}_N(S)$ is the essential image of

$$a^*[d] : \mathrm{EM}_N(\mathrm{Spec} k) \longrightarrow \mathrm{EM}_N(S).$$

Lemma 5.2 (Lowest algebraic direct image). *Let M be a vector bundle with integrable algebraic connection on the smooth d -fold S . In the perverse normalization of algebraic $\mathcal{D}$ -modules,*

$${}^p H^j(a_+ M) = 0 \quad (j < -d), \quad {}^p H^{-d}(a_+ M) = H_{\mathrm{dR}}^0(S, M) = H_{\nabla}^0(S, M).$$

Proof. The algebraic de Rham complex of M is concentrated in ordinary degrees $0, \dots, d$. The perverse normalization shifts a connection by d , giving the stated cohomological amplitude and identifying the lowest group with algebraic horizontal sections; see [27, Chapters 1 and 7]. $\square$

Put

$$C_N := {}^p H^{-d}(a_* N) \in \mathrm{EM}_N(\mathrm{Spec} k).$$

By operation compatibility and Lemma 5.2, the perverse cohomology of $a_* N$ vanishes below degree $-d$ after applying ρ_λ . Faithfulness reflects this vanishing to EM_N . Hence the canonical truncation map $C_N[d] \rightarrow a_* N$ and adjunction give

$$\epsilon_N : a^* C_N[d] \longrightarrow N.$$

Define

$$N^{\mathrm{fix}} := \mathrm{im}(\epsilon_N).$$

Lemma 5.3 (Horizontal evaluation). *Let $M = \rho_\lambda(N)$. The realization of ϵ_N is the evaluation map*

$$\mathcal{O}_S \otimes_k H_{\nabla}^0(S, M) \longrightarrow M.$$

Its image is the maximal trivial differential submodule of M .

Proof. By Lemma 5.2, the realization of C_N is $H_{\nabla}^0(S, M)$, and the adjunction morphism realizes as evaluation. Its image is trivial. Conversely, every trivial differential submodule has a basis of global algebraic horizontal sections and is therefore contained in the evaluation image. $\square$

Lemma 5.4 (Constant reflection). *Let $Q \in \text{EM}_N(S)$ be λ -lisse. If $\rho_\lambda(Q)$ is a trivial connection, then the adjunction evaluation*

$$a^*{}^p H^{-d}(a_* Q)[d] \longrightarrow Q$$

is an isomorphism. Thus Q is constant.

Proof. After realization the displayed map is the evaluation isomorphism

$$\mathcal{O}_S \otimes_k H_{\nabla}^0(S, \rho_\lambda(Q)) \xrightarrow{\sim} \rho_\lambda(Q).$$

Exact faithfulness of ρ_λ reflects this isomorphism. $\square$

Proposition 5.5 (Subquotient closure of constants). *Inside the λ -lisse heart, the constant subcategory is closed under subobjects and quotients, as well as tensor products and finite direct sums.*

Proof. A subobject or quotient of a trivial connection is trivial. Exactness of ρ_λ and Lemma 5.4 therefore show that every subobject or quotient of a constant λ -lisse object is constant. Smooth pullback preserves tensor products and finite direct sums. $\square$

Theorem 5.6 (Invariant exponential fixed part). *For the fixed scale λ ,*

$$\rho_\lambda(N^{\text{fix}}) = \text{Fix}(\rho_\lambda(N)).$$

Moreover, N^{fix} is constant and is the maximal constant exponential Nori subobject of N .

Proof. Exactness carries the image defining N^{fix} to the evaluation image, so the first assertion is Lemma 5.3. This realization is trivial; hence Lemma 5.4 shows that N^{fix} is constant.

Let $a^* B[d] \hookrightarrow N$ be any constant subobject. By adjunction it corresponds to a morphism $B[d] \rightarrow a_* N$. Since the perverse cohomology of $a_* N$ vanishes below degree $-d$, the morphism factors through $C_N[d] \rightarrow a_* N$. Its adjoint factors through ϵ_N , so the subobject is contained in N^{fix} . $\square$

6 Differential characters and Fourier geometry

6.1 Rank-one lifting

Fix a nonzero scale λ and recall $\rho_\lambda = \text{real}_\lambda \circ R_H^{\text{exp}}$ from Definition 4.6. Let $\mathcal{T} \subset \text{EM}_N(S)$ be a rigid Tannakian subcategory of λ -lisse objects, and let $\mathcal{T}_{\nabla}$ be the Tannakian subcategory of algebraic connections generated by $\rho_\lambda(\mathcal{T})$ and all its differential subquotients.

Definition 6.1 (Rank-one saturation). The pair $(\mathcal{T}, \rho_\lambda)$ is rank-one saturated if for every invertible $L \in \mathcal{T}_\nabla$ there is an invertible object $\mathcal{L}$ in the ambient relative exponential Nori category such that

$$\rho_\lambda(\mathcal{L}) \simeq L.$$

The lift may lie in the ambient category; the associated semi-invariant is recovered inside an object of $\mathcal{T}$ in Section 7.

6.2 A split Fourier-geometric criterion

We use the following sufficient condition.

Definition 6.2 (Split Fourier-geometric category). We call $(\mathcal{T}, \rho_\lambda)$ *split Fourier-geometric* if, for every $B \in \mathcal{T}_\nabla$, there exist a dense open $S^\circ \subset S$, a finite Galois cover $q : S' \rightarrow S^\circ$, and a good compactification such that:

(FG1) the Stokes-graded connection admits a global decomposition

$$\mathrm{gr}^{\mathrm{St}}(q^*B) \simeq \bigoplus_{\alpha \in A_B} \mathcal{E}^{g_\alpha} \otimes R_\alpha,$$

with $g_\alpha \in \mathcal{O}(S')$ pairwise distinct modulo constants and with R_α regular singular;

(FG2) every R_α is of geometric origin in the ordinary Nori category;

(FG3) the Galois descent datum on every rank-one summand $\mathcal{E}^{g_\alpha} \otimes A$ with A of finite order is effective in the relative exponential Nori category.

Lemma 6.3 (Rank-one selection of a Stokes grade). *Let L be a rank-one differential subquotient of a connection satisfying (FG1). Then, after the same finite cover,*

$$q^*L \simeq \mathcal{E}^{g_\alpha} \otimes A$$

for one α , where A is a rank-one regular-singular subquotient of R_α .

Proof. Good formal decomposition and the Stokes-graded functor are exact in the abelian category of Stokes-filtered local systems. A rank-one connection has one formal exponential factor and no off-diagonal Stokes operator. Therefore its graded object is a rank-one subquotient of one summand. Removing the selected exponential factor leaves a rank-one regular-singular subquotient of R_α . $\square$

Lemma 6.4 (Finite monodromy of the regular factor). *Under (FG2), the rank-one regular factor A in Lemma 6.3 has finite monodromy.*

Proof. Subquotients of ordinary geometric local systems are again of geometric origin. Lam–Litt prove that a rank-one complex local system on a smooth variety is of geometric origin if and only if it is torsion [24, Lemma 4.2.1]. Hence A has finite monodromy. Over the algebraically closed coefficient field, the corresponding character idempotent in the finite étale pushforward defines an invertible Artin object. $\square$

Theorem 6.5 (Generic split Fourier-geometric saturation). *Let $\mathcal{T}$ be a finitely generated split Fourier-geometric rigid tensor subcategory. After restriction to a suitable common dense open $S^\circ \subset S$, the pair $(\mathcal{T}|_{S^\circ}, \rho_\lambda)$ is rank-one saturated.*

Proof. Let $L \in \mathcal{T}_\nabla$ be invertible. By Lemmas 6.3 and 6.4, after a finite Galois cover

$$q^*L \simeq \mathcal{E}^g \otimes A$$

with g algebraic and A finite. At the fixed scale λ , the graph object with potential $-g/\lambda$ realizes $\mathcal{E}^g$ under the convention $\mathcal{E}^g = (\mathcal{O}, d - dg)$ and is invertible in the relative exponential Nori category. If m is the order of A , choose the finite étale cover which trivializes A . Over k , the character idempotent attached to A cuts out an invertible rank-one Artin object. The convolution tensor product of this Artin object with the graph object therefore realizes q^*L . Hypothesis (FG3) then produces an invertible object over S° realizing $L|_{S^\circ}$. Since $\mathcal{T}$ is finitely generated, the finitely many dense opens required for its generators and tensor constructions may be replaced by one common dense open. Hence the restricted category is rank-one saturated. $\square$

6.3 Nonzero Fourier fibers and nearby cycles

Let K be a regular geometric perverse-Nori object on $S \times \mathbb{A}_t^1$, and let

$$\widehat{K} = \mathrm{FL}_{t/S}(K)$$

be its partial Fourier–Laplace transform on $S \times \mathbb{A}_\lambda^1$. Fix $\lambda_0 \in \mathbb{C}^\times$ and shrink S so that

$$M_{\lambda_0} := i_{\lambda_0}^+ \widehat{K}$$

is a connection on S .

Lemma 6.6 (Finite graph reduction). *After shrinking S , there is a finite étale cover*

$$r : S' \longrightarrow S$$

and pairwise disjoint smooth algebraic graphs

$$H_i = \{t = u_i(s)\} \subset S' \times \mathbb{A}_t^1, \quad u_i \in \mathcal{O}(S'),$$

*such that the characteristic variety of the regular holonomic $\mathcal{D}$ -module underlying r^*K is contained in the union of the zero section and the conormal bundles $T_{H_i}^*(S' \times \mathbb{A}_t^1)$.*

Proof. Choose an algebraic Whitney stratification of $S \times \mathbb{A}_t^1$ adapted to the Betti realization of K . After deleting the images in S of all strata which do not dominate the base, every remaining non-open stratum has dimension $\dim S$: the ambient variety has dimension $\dim S + 1$, while a proper stratum dominating S has dimension at least $\dim S$ and codimension at least one. It is therefore generically finite over S . Normalize the finitely many such multisections, shrink once more so that they are finite étale, and pass to a common finite Galois refinement. They then split into smooth sections. Delete from S the Galois-stable union of the images of all pairwise intersections; after pulling back this open, the graphs are pairwise disjoint. The characteristic variety of a regular holonomic module constructible for the chosen Whitney stratification is contained in the union of the conormal varieties of the strata. The open stratum contributes the zero section, and the remaining strata contribute the conormals of the displayed graphs. $\square$

Lemma 6.7 (Nonzero-scale lissity). *After shrinking S further,*

$$\widehat{K}|_{S \times \mathbb{G}_m}$$

is a connection.

Proof. Apply Lemma 6.6 and work analytically on coordinate polydiscs of S' . Put $z = \lambda^{-1}$ on the punctured Fourier line. In the notation of Sabbah's parameterized partial Laplace transform, the characteristic-variety hypothesis of Appendix B, Proposition B.1 is now satisfied by the hypersurfaces $H_i = \{t - u_i = 0\}$. Sabbah states on p. 40, immediately before Proposition B.1, that the localized transform is a locally free $\mathcal{O}_{S'}[z, z^{-1}]$ -module of finite rank with an integrable connection. Proposition B.1 then identifies its formal logarithmic residual systems with the vanishing-cycle connections along the H_i . The microlocalization and decomposition inputs are the results of Douai–Sabbah cited there [32, Appendix B, p. 40 and Proposition B.1] [19, Propositions 1.18 and 1.20]. Under $z = \lambda^{-1}$ this localized transform is $r^*\widehat{K}|_{S' \times \mathbb{G}_m}$. Hence the latter is lisse. Lissity is local for the finite étale topology, so it descends from S' to S . $\square$

By Lemma 6.7, the Betti realization of $\widehat{K}$ on $S \times \mathbb{G}_m$ is, up to the standard perverse shift, a local system $\mathcal{G}$. It corresponds to a representation of

$$\pi_1(S \times \mathbb{G}_m) \simeq \pi_1(S) \times \mathbb{Z}.$$

We use perverse-normalized cycles

$${}^p\psi_f := R\psi_f[-1], \quad {}^p\phi_f := R\phi_f[-1].$$

Let $\mathcal{G}_\psi$ denote the unshifted local system on S underlying ${}^p\psi_\lambda \text{rat}(\widehat{K})$, equipped with its commuting nearby-monodromy operator T_λ .

Proposition 6.8 (Fixed fiber versus nearby cycles). *After forgetting T_λ , the local systems*

$$\mathcal{G}_{\lambda_0} \quad \text{and} \quad \text{For}_{T_\lambda}(\mathcal{G}_\psi)$$

are isomorphic, though not canonically so. Compatible basepoints and a path in $\mathbb{G}_m$ from λ_0 to a punctured neighbourhood of the origin determine such an isomorphism; two choices differ by a power of T_λ . Consequently, the fixed nonzero fiber and the nearby-cycle object have the same Jordan–Hölder constituents as $\pi_1(S)$ -representations.

Proof. By Lemma 6.7, $\mathcal{G}$ is a local system on $S \times \mathbb{G}_m$. Such a local system is a representation of $\pi_1(S) \times \mathbb{Z}$. Restriction to $S \times \{\lambda_0\}$ forgets the central $\mathbb{Z}$ -operator. After choosing compatible basepoints and a path to a punctured neighbourhood of the origin, perverse nearby cycles identify the same vector space and the same $\pi_1(S)$ -action, while retaining the commuting $\mathbb{Z}$ -operator as T_λ . Changing the path winds around the origin and composes the identification with a power of T_λ . Thus the underlying S -local systems are isomorphic, though not canonically so. $\square$

6.4 Sabbah's conventions and the relative comparison

Compactify the source variable by writing

$$X = S \times \mathbb{P}_t^1, \quad t' = t^{-1}$$

near infinity, and let

$$q : X \times \mathbb{A}_\lambda^1 \longrightarrow S \times \mathbb{A}_\lambda^1$$

be the projective projection. Let $\mathcal{M}$ be the regular holonomic $\mathcal{D}$ -module underlying the Hodge realization of K on $S \times \mathbb{A}_t^1$, and let

$$\mathcal{T} := \mathcal{M}_{\min}$$

be its regular minimal (middle) extension across $t = \infty$. Sabbah's Fourier kernel uses the localization $\mathcal{T}[*\infty] = \mathcal{M}[*\infty]$. Hence replacing an arbitrary regular extension by the minimal extension does not change the localized kernel, its partial Fourier–Laplace transform, or any nonzero Fourier fiber under study. He first constructs the Fourier kernel object $\mathcal{FT}$ on $X \times \mathbb{A}_\lambda^1$ and only then applies q_+ . Because the Fourier kernel depends only on $\mathcal{T}[*\infty]$, the normalized classical partial Fourier–Laplace transform is the usual exact transform of the localized regular holonomic perverse module. Its projective direct image is therefore concentrated in perverse degree zero:

$${}^p\mathcal{H}^j q_+(\mathcal{FT}) = 0 \quad (j \neq 0), \quad \widehat{\mathcal{T}} := {}^p\mathcal{H}^0 q_+(\mathcal{FT}),$$

and $\widehat{\mathcal{T}}$ underlies $\widehat{K}$; see [36, Appendix A] and [27, Chapter 3]. The unipotent case is treated before projective direct image.

We keep Sabbah's nearby-cycle exponent labels and write his Fourier coordinate τ as λ . With the perverse normalization above, his comparison has the following form on underlying regular $\mathcal{D}$ -modules and local systems:

label	comparison before q_+
$-1 \leq \operatorname{Re} \alpha < 0, \alpha \neq -1$	$\Psi_{\lambda, \alpha}(\mathcal{FT}) \simeq i_{\infty, +} \Psi_{t', \alpha}(\mathcal{T})$
0	$\phi_{\lambda, 0}(\mathcal{FT}) \simeq i_{\infty, +} \psi_{t', -1}(\mathcal{T})$
-1	$0 \rightarrow i_{\infty, +} \ker N_{t'} \rightarrow \ker N_\lambda \rightarrow \mathcal{T} \rightarrow 0.$

(6.1)

The first two lines are Proposition 5.8; the first two comparisons after projective direct image are Corollary 5.20 [31, Proposition 5.8 and Corollary 5.20]. The last exact sequence is also part of Proposition 5.8. Here $\phi_{\lambda, 0}$ is the vanishing-cycle object for eigenvalue 1, whereas the unipotent nearby-cycle object is $\psi_{\lambda, -1}$.

Two twistor normalizations in Sabbah's formulas do not alter the underlying $\mathcal{D}$ -module or local system. By Sabbah's Lemma 5.6, the factor $\mathcal{O}_{\Omega_0}(-D_\alpha)$ is absorbed at the level of R -triples in the comparison above. The half twist $(-1/2)$ rescales the twistor pairing and weight normalization. Neither is a Nori half-Tate twist, so both are omitted when discussing simple local-system constituents [31, Lemma 5.6].

Theorem 6.9 (Nearby-cycle geometricity). *Every simple lisse constituent of*

$${}^p\psi_{\lambda=0}(\widehat{K})$$

is of ordinary geometric origin.

Proof. We first treat a simple pure constituent of the Hodge realization of K . Its regular minimal extension $\mathcal{T} = \mathcal{M}_{\min}$ underlies a polarizable regular twistor $\mathcal{D}$ -module by the Hodge-to-twistor realization [30, Section 5]. Sabbah's relative Fourier comparison then applies with parameter space $Y = S$ [31].

For the non-unipotent labels, the first line of (6.1), followed by the corresponding comparison after q_+ , identifies every simple constituent of the Fourier nearby-cycle object with a constituent of an ordinary nearby-cycle object at $t = \infty$. The second line treats the eigenvalue-one vanishing-cycle object.

For the unipotent nearby-cycle object

$$V = {}^p\psi_{\lambda, -1}(\mathcal{FT}), \quad N = N_\lambda,$$

use the finite filtration

$$0 \subset \ker N \subset \ker N^2 \subset \cdots \subset V.$$

For each $j \geq 1$, the map induced by N^{j-1} embeds $\ker N^j / \ker N^{j-1}$ into $\ker N$. Hence every simple constituent of V occurs in $\ker N$. Sabbah's pre- q_+ exact sequence expresses $\ker N$ as an extension of $\mathcal{T}$ by $i_{\infty,+} \ker N_{t'}$. Therefore every simple constituent of V occurs in

$$\mathcal{T} \quad \text{or} \quad i_{\infty,+} {}^p\psi_{t',-1}(\mathcal{T}). \quad (6.2)$$

Now apply the derived proper direct image Rq_* . Proper compatibility of nearby cycles identifies Rq_*V with the unipotent nearby-cycle complex of $q_+(\mathcal{FT})$. Applying Rq_* to a finite constituent filtration of V gives distinguished triangles whose long exact perverse-cohomology sequences show that every simple constituent of ${}^p\psi_{\lambda,-1}(\widehat{K})$ occurs among the perverse-cohomology constituents of the proper direct images of the two ordinary geometric objects in (6.2). Thus Sabbah's unipotent sequence is used only before projective direct image.

Middle extension, nearby cycles, proper direct image, and perverse cohomology are operations in the ordinary perverse-Nori coefficient system [16, 35]. Hence every constituent produced above is of ordinary Nori-geometric origin.

For mixed K , work on its mixed Hodge realization

$$H := R_H(K) \in \text{MHM}(S \times \mathbb{A}_t^1).$$

Saito's weight filtration is strict, and normalized partial Fourier transform and perverse nearby cycles are exact. They therefore induce a finite filtration on ${}^p\psi_{\lambda}(\text{FL}_{t/S} H)$ whose graded subquotients are subquotients of

$${}^p\psi_{\lambda} \text{FL}_{t/S}(\text{gr}_m^W H).$$

Each pure graded object is semisimple. Its simple strict-support constituents belong, on the underlying local-system side, to the abelian subquotient-closed category of local systems of geometric origin identified by Jacobsen–Terenzi [20, Corollary 5.15]. The pure argument above applies constituent by constituent. Hence every simple constituent of the mixed nearby-cycle object is of geometric origin [29]. Sabbah's comparison is used only to identify complex isomorphism classes; geometric origin is supplied by the ordinary geometric objects on the right. $\square$

Remark 6.10 (The two-graph family). For the family $y^2 = t(t-s)$, the local system extends at $t = \infty$, so ${}^p\psi_{t',-1}\mathcal{T}$ has rank one, $N_{t'} = 0$, and trivial s -monodromy. Moreover $H^0(q_+\mathcal{T}) = H^2(q_+\mathcal{T}) = 0$ because there are no invariant or coinvariant vectors, while $\chi(\mathbb{P}^1 \setminus \{0, s\}, \mathcal{L}) = 0$; hence $q_+\mathcal{T} = 0$. Sabbah's upstream exact sequence gives $\dim \ker N_{\lambda} = 1$, whereas the unipotent nearby-cycle object has rank two. Thus N_{λ} has a single Jordan block and the two rank-one Jordan–Hölder constituents are trivial, as in the confluent system $M(1/2, 1, \lambda s)$.

Corollary 6.11 (Finite monodromy of rank-one Fourier factors). *Every rank-one Jordan–Hölder constituent of a fixed nonzero Fourier fiber has finite Betti monodromy.*

Proof. By Proposition 6.8, after forgetting the nearby-monodromy operator the fixed fiber and the nearby-cycle object have the same underlying $\pi_1(S)$ -representation. By Theorem 6.9, the resulting rank-one constituent lies in the Nori category of local systems of geometric origin. Jacobsen–Terenzi identify this category with Ayoub's category of local systems of geometric origin [20, Corollary 5.15]. Lam–Litt prove that a rank-one complex local system on a smooth variety is of geometric origin if and only if it is torsion [24, Lemma 4.2.1]. $\square$

6.5 The arithmetic exponential kernel

From this point $k = \overline{\mathbb{Q}}$, and we use the convention $\mathcal{E}^f = (\mathcal{O}_U, d - df)$.

Proposition 6.12 (Picard 1-motive convention). *Let U be smooth and let $\overline{U}$ be a smooth proper compactification with normal-crossings boundary D . With the covariance conventions used in this paper,*

$$T_H(\mathrm{Pic}^+(U)) \simeq H^1(U, \mathbb{Z}(1)), \quad \mathrm{Pic}^+(U) = [\mathrm{Div}_D^0(\overline{U}) \longrightarrow \mathrm{Pic}^0(\overline{U})].$$

A morphism $\mathbb{Q}(0) \rightarrow \mathrm{Pic}^+(U)_{\mathbb{Q}}$ is represented by a rational boundary divisor in the kernel of the map to $\mathrm{Pic}^0(\overline{U}) \otimes \mathbb{Q}$. After multiplication by an integer making the divisor principal, say $\mathrm{div}(u)$, its de Rham realization is the logarithmic class $[d \log u]$.

Proof. This is the cohomological Picard 1-motive and its Hodge and de Rham realizations as constructed by Barbieri–Viale and Srinivas; see [6, Sections 2–4] and [17, Section 10]. $\square$

Theorem 6.13 (Arithmetic exponential-kernel theorem). *Let $U/\overline{\mathbb{Q}}$ be smooth and connected, and let L be a rank-one algebraic integrable connection over $\overline{\mathbb{Q}}$. If its Betti local system has finite monodromy, then there exist a dense affine open immersion $j : U^\circ \hookrightarrow U$, a function $f \in \mathcal{O}(U^\circ)$, and a finite-order rank-one algebraic connection A on U° such that*

$$j^*L \simeq \mathcal{E}^f \otimes A.$$

Proof. Let A_0 be the finite Artin connection with the same Betti character as L ; it exists over $\overline{\mathbb{Q}}$ by invariance of finite étale covers under extension of algebraically closed fields. Set $L_0 = L \otimes A_0^{-1}$, so that L_0 has trivial Betti monodromy. Choose a dense affine open immersion $j : U^\circ \hookrightarrow U$ on which the underlying line bundle of L_0 is trivial, and replace U, L_0 by U°, j^*L_0 for the remainder of the proof. Write

$$\nabla = d + \omega, \quad \omega \in \Gamma(U, \Omega_{U/\overline{\mathbb{Q}}}^1), \quad d\omega = 0.$$

Trivial monodromy gives

$$\int_\gamma \omega \in 2\pi i \mathbb{Z} \quad (\gamma \in H_1(U(\mathbb{C}), \mathbb{Z})).$$

Let $\eta \in H_{\mathrm{B}}^1(U, \mathbb{Q})$ be the integral class defined by

$$\langle \eta, \gamma \rangle = (2\pi i)^{-1} \int_\gamma \omega.$$

Then comparison sends $[\omega]_{\mathrm{dR}}$ to $2\pi i \eta$. If $\eta = 0$, comparison gives $[\omega]_{\mathrm{dR}} = 0$; since U is affine there is $F \in \mathcal{O}(U)$ with $\omega = dF$, and $L_0 \simeq \mathcal{E}^{-F}$. We henceforth assume $\eta \neq 0$.

The de Rham–Betti object $H_{\mathrm{dRB}}^1(U)$ then contains the one-dimensional subobject determined by the specific pair $([\omega]_{\mathrm{dR}}, \eta_{\mathrm{B}})$:

$$\mathbb{Q}(-1) \hookrightarrow H_{\mathrm{dRB}}^1(U), \tag{6.3}$$

where $\mathbb{Q}(-1)$ is the Tate object whose period is $2\pi i$. Twisting by $\mathbb{Q}(1)$ gives

$$\mathbb{Q}(0) \hookrightarrow H_{\mathrm{dRB}}^1(U)(1). \tag{6.4}$$

Ayoub–Barbieri–Viale identify the degree-one Nori category with Deligne 1-motives with torsion [5]. By Proposition 6.12, the cohomological Picard 1-motive of U is

$$\mathrm{Pic}^+(U) = [\mathrm{Div}_D^0(\overline{U}) \longrightarrow \mathrm{Pic}^0(\overline{U})], \tag{6.5}$$

and realizes $H^1(U, \mathbb{Z}(1))$. Huber–Wüstholz prove that the de Rham–Betti realization of 1-motives is fully faithful and that its image is closed under subquotients [13, Theorem 9.14]. Closure under subquotients shows that the specific Tate line in (6.4) is the realization of a 1-motivic subobject; full faithfulness lifts its inclusion to the specific morphism

$$\mathbb{Q}(0) \longrightarrow \mathrm{Pic}^+(U)_{\mathbb{Q}}$$

whose realization is (6.4).

In the isogeny category this morphism is represented by a rational boundary divisor $\Delta \in \mathrm{Div}_D^0(\overline{U}) \otimes \mathbb{Q}$ whose class in $\mathrm{Pic}^0(\overline{U}) \otimes \mathbb{Q}$ is zero. Choose an integer $N > 0$ which both clears denominators and kills the resulting torsion Picard class. Then the integral divisor $N\Delta$ has trivial associated line bundle, so

$$N\Delta = \mathrm{div}(u)$$

for a rational function whose zeros and poles lie in D ; equivalently, $u \in \mathcal{O}(U)^\times$. By Proposition 6.12, the lattice generator represented by the principal boundary divisor $\mathrm{div}(u)$ maps under de Rham realization to the logarithmic class $[d \log u]$. Compatibility of the lifted morphism with de Rham realization therefore gives

$$N[\omega] = [d \log u] \quad \text{in } H_{\mathrm{dR}}^1(U/\overline{\mathbb{Q}}).$$

Since U is affine, algebraic de Rham cohomology in degree one is computed by global forms [26]. Hence

$$N\omega - d \log u = dF$$

for some $F \in \mathcal{O}(U)$. Therefore

$$L_0 \simeq \mathcal{E}^{-F/N} \otimes \left(\mathcal{O}_U, d + \frac{1}{N} d \log u \right).$$

The second factor has order dividing N , since its N th tensor power is trivialized by u^{-1} . Tensoring back by A_0 proves the theorem. $\square$

Remark 6.14 (Arithmetic scope). Over $\mathbb{C}$, a topologically trivial rank-one algebraic connection can carry a nonzero second-kind de Rham class and need not be an exponential times a finite connection. Huber–Wüstholz excludes this non-Tate component over $\overline{\mathbb{Q}}$. For example, if $U = E \setminus \{0\}$ for an elliptic curve, then $\mathrm{Div}_{\{0\}}^0(E) = 0$, so $\mathrm{Pic}^+(U)$ has no toric $\mathbb{Q}(0)$ -subobject. A nonzero second-kind class over $\mathbb{C}$ therefore cannot arise from the Tate-line mechanism used in the proof.

6.6 Generic saturation for the Kontsevich–Zagier category

Let $\eta = \mathrm{Spec} k(S)$ be the generic point. We write

$$\mathrm{EM}_N(\eta) := \varinjlim_{U \subset S} \mathrm{EM}_N(U),$$

where objects and morphisms are identified after restriction to a smaller nonempty smooth open. Restriction is exact and tensor-compatible, so every finite diagram in this filtered 2-colimit has a common model and the generic realization is an exact faithful tensor functor to algebraic differential modules over $k(S)$.

Let $\mathcal{T}_{\mathrm{KZ}}(\eta)$ denote the filtered union of all finitely generated rigid Tannakian subcategories of $\mathrm{EM}_N(\eta)$ generated by dualizable restrictions of elementary relative Kontsevich–Zagier exponential data.

Theorem 6.15 (Generic rank-one saturation). *For every $\lambda \in \overline{\mathbb{Q}}^\times$, every finitely generated rigid Tannakian subcategory of $\mathcal{T}_{\text{KZ}}(\eta)$ is rank-one saturated.*

Proof. Let B be a tensor construction and subquotient in such a category, and let L be an invertible differential subquotient of $\rho_\lambda(B)$. Choose a common smooth open model on which B and L are lisse and choose a regular geometric pre-Fourier representative K_B . Such a representative exists because the quotient functor defining $\text{EM}_\mathbb{N}$ is essentially surjective on the heart; changing it by a constant object does not change any nonzero exponential realization. Additive convolution and the six operations preserve regular geometric origin.

After a further dense-open restriction, Corollary 6.11 shows that the Betti local system of L has finite monodromy. Applying Theorem 6.13 on a dense affine open gives

$$L \simeq \mathcal{E}^f \otimes A$$

with f algebraic and A finite order. At the fixed scale λ , the graph object with potential $-f/\lambda$ realizes $\mathcal{E}^f$. The finite-order character A is cut out by its character idempotent in the finite étale pushforward which trivializes it. Their convolution product is an invertible object in the generic exponential Nori category realizing L . Hence L has an ambient invertible lift. $\square$

Remark 6.16 (Models on dense opens). Every finite packet is defined on a common dense lisse open. If a rank-one connection is defined on a larger smooth base, finite monodromy on a dense open implies finite monodromy on the base because $\pi_1(U^{\text{an}}) \twoheadrightarrow \pi_1(S^{\text{an}})$.

7 Twisted fixed parts and Galois exactness

Fix $\lambda \in \overline{\mathbb{Q}}^\times$ and let $\mathcal{T}$ be a finitely generated rigid Tannakian subcategory of $\mathcal{T}_{\text{KZ}}(\eta)$. Choose a common smooth connected open model S on which every generator is λ -lisse; all tensor constructions are then λ -lisse as well. Fix a fiber functor at a geometric point $s \in S(\mathbb{C})$ and write

$$G_{\text{mot}} = \text{Aut}^\otimes(\omega_s|_{\mathcal{T}})$$

for the motivic Galois group. Let $\mathcal{T}_\nabla$ be the differential tensor category generated by $\rho_\lambda(\mathcal{T})$ and write

$$G_{\text{diff}} = \text{Aut}^\otimes(\omega_s|_{\mathcal{T}_\nabla}).$$

The exact faithful tensor realization gives a closed immersion $G_{\text{diff}} \hookrightarrow G_{\text{mot}}$.

7.1 Semi-invariant spaces

For a G_{mot} -representation V and a character $\chi : G_{\text{diff}} \rightarrow \mathbb{G}_m$, put

$$V^\chi = \{v \in V : hv = \chi(h)v \text{ for all } h \in G_{\text{diff}}\}.$$

Proposition 7.1 (Motivic semi-invariants). *Assume rank-one saturation. For every $B \in \mathcal{T}$ and every character χ , the subspace $\rho_\lambda(B)^\chi$ is the realization of a subobject of B belonging to $\mathcal{T}$.*

Proof. Let L_χ be the invertible differential object associated with χ and choose an ambient invertible lift $\mathcal{L}_\chi$ by rank-one saturation. Apply Theorem 5.6 to the ambient object $B \otimes \mathcal{L}_\chi^{-1}$ and let

$$\mathcal{C}_\chi := (B \otimes \mathcal{L}_\chi^{-1})^{\text{fix}}.$$

Its realization is

$$(\rho_\lambda(B) \otimes L_\chi^{-1})^{G_{\text{diff}}}.$$

Tensor the inclusion $\mathcal{C}_\chi \hookrightarrow B \otimes \mathcal{L}_\chi^{-1}$ by $\mathcal{L}_\chi$ and define

$$\mathcal{W}_\chi := \text{im}(\mathcal{L}_\chi \otimes \mathcal{C}_\chi \longrightarrow B).$$

Exactness of realization gives

$$\rho_\lambda(\mathcal{W}_\chi) = L_\chi \otimes (\rho_\lambda(B) \otimes L_\chi^{-1})^{G_{\text{diff}}} = \rho_\lambda(B)^\chi.$$

Although $\mathcal{L}_\chi$ was chosen in the ambient category, $\mathcal{W}_\chi$ is a subobject of B . The Tannakian category generated by B is closed under subobjects, so $\mathcal{W}_\chi \in \mathcal{T}$. $\square$

7.2 Normality

We use Andre's criterion in the following form.

Lemma 7.2 (Andre's normality criterion). *Let $H \subseteq G$ be affine algebraic groups over a field of characteristic zero. If, for every finite-dimensional G -representation V and every character χ of H , the semi-invariant space V^χ is G -stable, then H is normal in G .*

Proof. See [2] and the Tannakian formulation in [25]. $\square$

Theorem 7.3 (Relative exponential Galois exactness). *Assume rank-one saturation. Let $\mathcal{T}_0 \subseteq \mathcal{T}$ be the full subcategory of constant objects and let G_{const} be its Tannakian group. Then*

$$1 \longrightarrow G_{\text{diff}} \longrightarrow G_{\text{mot}} \longrightarrow G_{\text{const}} \longrightarrow 1$$

is exact.

Proof. By Proposition 7.1 and Lemma 7.2, G_{diff} is normal in G_{mot} . By Proposition 5.5, $\mathcal{T}_0$ is a full tensor subcategory stable under subquotients; hence its inclusion induces a faithfully flat morphism $G_{\text{mot}} \twoheadrightarrow G_{\text{const}}$ [18, Proposition 2.21].

An object of $\mathcal{T}$ is trivial under G_{diff} if and only if its λ -differential realization is trivial. By Lemma 5.4, such an object is constant. Conversely, constant objects have trivial relative differential realization. Therefore the Tannakian quotient corresponding to $G_{\text{mot}}/G_{\text{diff}}$ is $\mathcal{T}_0$, proving exactness. $\square$

Corollary 7.4 (Generic split Fourier-geometric exactness). *If $\mathcal{T}$ is finitely generated and split Fourier-geometric, then after restriction to a suitable common dense open the exact sequence of Theorem 7.3 holds.*

8 The relative functional period theorem

8.1 Comparison torsors and the constant-period base

Let $K = k(S)$ and let ω_{dR} and ω_{B} denote the de Rham and rapid-decay Betti fiber functors on $\mathcal{T}$. The relative motivic comparison torsor is

$$\mathcal{X}_{\text{mot}} = \text{Isom}^\otimes(\omega_{\text{B}} \otimes_k K, \omega_{\text{dR}}),$$

and its restriction to the constant subcategory is denoted $\mathcal{X}_0$.

Let p_0 be the actual constant comparison point and let $Z_0 \subseteq \mathcal{X}_0$ be its Zariski closure. Put

$$A_0 := \mathcal{O}(Z_0), \quad R_0 := K \otimes_k A_0, \quad F_0 := \text{Frac}(A_0), \quad K_0 := \text{Frac}(R_0).$$

The kernel of evaluation at p_0 is prime, so A_0 is a domain. Since S is geometrically integral over the algebraically closed field k , R_0 is a domain as well. We extend the derivations of K to K_0 by declaring F_0 to be constant.

Define

$$\mathcal{X}_{\text{rel}} := \mathcal{X}_{\text{mot}} \times_{\mathcal{X}_0} Z_0.$$

The algebra A_0 is the coordinate ring of the Zariski closure of the actual constant comparison point.

Proposition 8.1 (Relative torsor). *Under rank-one saturation, the morphism*

$$\mathcal{X}_{\text{rel}} \longrightarrow \text{Spec } R_0$$

is a G_{diff, R_0} -torsor. Consequently $\mathcal{O}(\mathcal{X}_{\text{rel}})$ is faithfully flat and torsion-free over R_0 .

Proof. By Theorem 7.3, restriction of comparison isomorphisms to the constant subcategory is the quotient of the G_{mot} -torsor $\mathcal{X}_{\text{mot}}$ by the normal subgroup G_{diff} . Thus $\mathcal{X}_{\text{mot}} \rightarrow \mathcal{X}_0$ is a G_{diff} -torsor. Base change along $Z_0 \rightarrow \mathcal{X}_0$, followed by the scalar extension from k to K , gives the asserted torsor over R_0 . Torsors under affine group schemes are faithfully flat over their base. $\square$

8.2 Comparison with Picard–Vessiot theory

Let $\mathcal{X}_{\text{PV}}$ be the Picard–Vessiot torsor of the differential tensor category $\mathcal{T}_{\nabla}$ over the differential field K_0 , whose field of constants contains F_0 . On a simply connected analytic open, the comparison matrices of objects of $\mathcal{T}$ generate the Picard–Vessiot algebra. Evaluation of motivic matrix coefficients at the same matrices gives a G_{diff, K_0} -equivariant closed immersion

$$\mathcal{X}_{\text{PV}} \longrightarrow \mathcal{X}_{\text{rel}} \times_{R_0} K_0.$$

Lemma 8.2 (Equivariant maps of torsors). *An equivariant morphism between two nonempty torsors under the same affine group over a field is an isomorphism.*

Proof. After a faithfully flat extension trivializing both torsors, the morphism becomes an equivariant endomorphism of the group acting on itself by translation, hence a translation and therefore an isomorphism. Descent gives the result over the original field. $\square$

Theorem 8.3 (Relative functional exponential-period theorem). *Assume rank-one saturation. The generic fiber of $\mathcal{X}_{\text{rel}}$ is the Picard–Vessiot torsor. The analytic relative period map*

$$\text{per}_{S, \text{rel}}^{\text{exp}} : \mathcal{O}(\mathcal{X}_{\text{rel}}) \longrightarrow \mathcal{M}(\tilde{S})$$

is injective.

Proof. By Proposition 8.1, the generic fiber $\mathcal{X}_{\text{rel}} \times_{R_0} K_0$ and $\mathcal{X}_{\text{PV}}$ are nonempty torsors under the same group. The preceding closed immersion is therefore an isomorphism by Lemma 8.2.

Let $x \in \mathcal{O}(\mathcal{X}_{\text{rel}})$ map to zero analytically. After tensoring with K_0 , the torsor isomorphism identifies x with an element of the Picard–Vessiot algebra which is zero in the meromorphic solution field; hence $x \otimes 1 = 0$. Thus x is R_0 -torsion. Faithful flatness in Proposition 8.1 makes $\mathcal{O}(\mathcal{X}_{\text{rel}})$ torsion-free over R_0 , so $x = 0$. $\square$

Corollary 8.4 (Functional relation interpretation). *Under rank-one saturation, every polynomial functional relation among the relative exponential periods of $\mathcal{T}$, with coefficients in the actual constant exponential-period field, is induced by a tensor relation in $\mathcal{T}$.*

Proof. Polynomial expressions in matrix coefficients are matrix coefficients of finite tensor constructions. Injectivity of the comparison algebra therefore identifies the kernel of functional evaluation with the tensor relations defining the comparison torsor. $\square$

Remark 8.5 (Categorical and elementary period algebras). Corollary 8.4 gives completeness in the motivic formal-period algebra. It does not by itself prove that this algebra has a minimal presentation solely by linearity, elementary algebraic changes of variables, Fubini, and twisted Stokes. The comparison between the categorical presentation and a chosen elementary Kontsevich–Zagier presentation is a separate theorem.

9 Relation with Kontsevich–Zagier exponential periods

Let $\mathcal{T}_{\text{KZ}}(\eta)$ be the generic relative exponential Nori Tannakian envelope introduced before Theorem 6.15. It is generated by the restrictions of elementary data

$$(X \rightarrow S, Y, f, n, i), \quad f : X \rightarrow \mathbb{A}^1 \text{ regular},$$

and their tensor operations and duals. Every finite packet has a common dense lisse model. The comparison theorems for exponential periods identify its matrix coefficients with cohomological rapid-decay period functions [9, 10, 11].

Theorem 9.1 (Relative exponential Nori–Ayoub theorem). *For the canonical realization $\lambda = 1$, the generic relative formal exponential-period map of the filtered Kontsevich–Zagier exponential Tannakian envelope is injective after the algebraic relations among constant exponential periods have been imposed. The same statement holds for every $\lambda \in \overline{\mathbb{Q}}^\times$. Equivalently, every algebraic functional relation among a finite packet of relative exponential periods is induced, on a common dense lisse open, by a tensor relation in the relative exponential Nori category.*

Proof. Every finite packet is contained in a finitely generated rigid Tannakian subcategory of $\mathcal{T}_{\text{KZ}}(\eta)$. By Theorem 6.15, that subcategory is rank-one saturated at the chosen algebraic nonzero scale. Choose a common dense lisse model and apply Theorem 8.3. Taking the filtered colimit over finite packets proves the generic statement. Since analytic functions agreeing on a dense open agree wherever both sides are defined, this is the asserted functional relation theorem. $\square$

9.1 Arithmetic specialization

Theorem 9.1 does not solve the numerical exponential-period conjecture. It controls identities valid identically in the parameters. A relation created at an algebraic fiber requires an arithmetic lifting theorem. In the E -function setting this role is played by Beukers’ refinement of the Siegel–Shidlovskii theorem; in period-valued, Gamma, and more general irregular settings the corresponding specialization theorems remain independent problems. The cyclotomic Gamma calculation in Section 10.2 makes this separation explicit.

10 Applications and examples

10.1 Polynomial exponential integrals

For a finite packet of integrals

$$F_j(z) = \int_{\gamma_j} p_j(x) e^{-z q_j(x)} dx,$$

the Fourier support is finite and algebraic. Whenever the companion polynomial-line analysis verifies that a finite cover splits the critical-value support and that the regular coefficient systems have finite monodromy, the generic split Fourier-geometric theorem applies. The present result then gives the categorical completeness of functional relations; specialization at an algebraic value of z remains the separate arithmetic E -function step.

10.2 Cyclotomic Gamma packets and the Rohrlich–Lang boundary

The main theorem gives categorical completeness for these rapid-decay periods. For the standard one-parameter Gamma packet, the complete functional ideal can also be computed directly. The resulting Gamma-specific presentation is elementary and, to our knowledge, has not been recorded in this form.

Fix an integer $N \geq 2$ and distinct integers

$$1 \leq m_1 < \cdots < m_s < N.$$

Put

$$r_j = \frac{m_j}{N}, \quad \gamma_j = \Gamma(r_j), \quad C_\Gamma = \overline{\mathbb{Q}}(\gamma_1, \dots, \gamma_s) \subset \mathbb{C}.$$

For $\Re z > 0$, define

$$G_j(z) = N \int_0^\infty x^{m_j-1} e^{-zx^N} dx,$$

and continue the function to the universal cover $\widetilde{\mathbb{G}_m}$. For a multi-index $\mathbf{a} = (a_1, \dots, a_s) \in \mathbb{N}^s$, write

$$X^{\mathbf{a}} = \prod_{j=1}^s X_j^{a_j}, \quad \gamma^{\mathbf{a}} = \prod_{j=1}^s \gamma_j^{a_j}, \quad M(\mathbf{a}) = \sum_{j=1}^s a_j m_j.$$

Theorem 10.1 (Functional cyclotomic Gamma presentation). *One has*

$$G_j(z) = \gamma_j z^{-m_j/N}.$$

Let

$$\Phi : C_\Gamma(z)[X_1, \dots, X_s] \longrightarrow \mathcal{M}(\widetilde{\mathbb{G}_m}), \quad X_j \longmapsto G_j(z).$$

Then

$$\ker \Phi = \left(\gamma^{\mathbf{b}} X^{\mathbf{a}} - z^{-q} \gamma^{\mathbf{a}} X^{\mathbf{b}} : M(\mathbf{a}) - M(\mathbf{b}) = qN \right), \quad (10.1)$$

where $\mathbf{a}, \mathbf{b} \in \mathbb{N}^s$ and $q \in \mathbb{Z}$. Thus every polynomial functional relation in the packet is generated by the Kummer-weight binomials in (10.1).

Proof. For positive real z , the substitution $u = zx^N$ gives

$$G_j(z) = z^{-m_j/N} \int_0^\infty u^{m_j/N-1} e^{-u} du = \gamma_j z^{-m_j/N}.$$

Both sides are holomorphic on the right half-plane, so the identity holds there and then on $\widetilde{\mathbb{G}_m}$ by analytic continuation.

Choose the global function

$$t = z^{-1/N}$$

on $\widetilde{\mathbb{G}_m}$. Then $t^N = z^{-1}$ and

$$G_j(z) = \gamma_j t^{m_j}.$$

Since $\mu_N \subset \overline{\mathbb{Q}} \subset C_\Gamma$, the cyclic action $t \mapsto \zeta t$ has fixed field $C_\Gamma(t^N) = C_\Gamma(z)$. Hence

$$[C_\Gamma(t) : C_\Gamma(z)] = N, \quad 1, t, \dots, t^{N-1}$$

is a basis over $C_\Gamma(z)$.

Let I denote the ideal on the right-hand side of (10.1). Every displayed generator maps to zero, since $M(\mathbf{a}) - M(\mathbf{b}) = qN$ implies

$$t^{M(\mathbf{a})} = z^{-q} t^{M(\mathbf{b})}.$$

Thus $I \subseteq \ker \Phi$.

Conversely, let

$$P = \sum_{\mathbf{a}} c_{\mathbf{a}}(z) X^{\mathbf{a}} \in \ker \Phi.$$

Write uniquely

$$M(\mathbf{a}) = q_{\mathbf{a}}N + \rho_{\mathbf{a}}, \quad 0 \leq \rho_{\mathbf{a}} < N.$$

Then

$$0 = \Phi(P) = \sum_{\rho=0}^{N-1} t^\rho \left(\sum_{\rho_{\mathbf{a}}=\rho} c_{\mathbf{a}}(z) \gamma^{\mathbf{a}} z^{-q_{\mathbf{a}}} \right).$$

Linear independence of $1, t, \dots, t^{N-1}$ gives, for every ρ ,

$$\sum_{\rho_{\mathbf{a}}=\rho} c_{\mathbf{a}}(z) \gamma^{\mathbf{a}} z^{-q_{\mathbf{a}}} = 0. \tag{10.2}$$

Fix one exponent vector $\mathbf{a}_0$ in a nonempty residue class. Modulo I ,

$$X^{\mathbf{a}} \equiv z^{-(q_{\mathbf{a}}-q_{\mathbf{a}_0})} \frac{\gamma^{\mathbf{a}}}{\gamma^{\mathbf{a}_0}} X^{\mathbf{a}_0} \quad (\rho_{\mathbf{a}} = \rho_{\mathbf{a}_0}).$$

Therefore the part of P in this residue class is congruent modulo I to

$$\frac{z^{q_{\mathbf{a}_0}} X^{\mathbf{a}_0}}{\gamma^{\mathbf{a}_0}} \left(\sum_{\rho_{\mathbf{a}}=\rho_{\mathbf{a}_0}} c_{\mathbf{a}}(z) \gamma^{\mathbf{a}} z^{-q_{\mathbf{a}}} \right),$$

which vanishes by (10.2). Summing over the residue classes gives $P \in I$. $\square$

For the complete packet, the general binomial ideal has a particularly small set of generators.

Corollary 10.2 (Complete fixed-denominator packet). *For*

$$G_m(z) = \Gamma(m/N)z^{-m/N}, \quad 1 \leq m < N,$$

put $\gamma_m = \Gamma(m/N)$. The kernel of

$$C_{\Gamma,N}(z)[X_1, \dots, X_{N-1}] \longrightarrow \mathcal{M}(\widetilde{\mathbb{G}_m}), \quad X_m \longmapsto G_m(z),$$

where

$$C_{\Gamma,N} = \overline{\mathbb{Q}}(\gamma_1, \dots, \gamma_{N-1}),$$

is generated by

$$\gamma_1^m X_m - \gamma_m X_1^m \quad (2 \leq m < N)$$

and

$$X_1^N - \gamma_1^N z^{-1}.$$

Proof. The first family eliminates every X_m in favor of X_1 . The resulting quotient is

$$C_{\Gamma,N}(z)[X_1]/(X_1^N - \gamma_1^N z^{-1}).$$

The map $X_1 \mapsto \gamma_1 t$ embeds this quotient into $C_{\Gamma,N}(t)$ because $1, t, \dots, t^{N-1}$ are linearly independent over $C_{\Gamma,N}(z)$. Hence the displayed relations generate the full kernel. $\square$

For example, when $N = 3$,

$$G_1(z) = \Gamma(1/3)z^{-1/3}, \quad G_2(z) = \Gamma(2/3)z^{-2/3},$$

and the complete functional ideal is generated by

$$\Gamma(1/3)^2 G_2(z) - \Gamma(2/3) G_1(z)^2$$

and

$$G_1(z)^3 - \Gamma(1/3)^3 z^{-1}.$$

Remark 10.3 (Relation with the Rohrlich–Lang conjectures). Rohrlich’s conjecture predicts that the multiplicative algebraic relations among special rational Gamma values and $2\pi i$ are generated by the standard translation, reflection, and multiplication identities. Lang’s strengthening makes the corresponding prediction for all polynomial algebraic relations [1, §1.4]; see also [28]. Koblitz and Ogus proved the algebraicity implication in the Deligne–Koblitz–Ogus criterion for Gamma monomials; the conjectural converse is a form of Rohrlich’s conjecture [23].

Theorem 10.1 is a relative functional statement, not a result on those constant-value conjectures. Its coefficient field C_Γ is the actual field generated by the chosen Gamma constants, so all relations among those constants have already been imposed. Before extending scalars from $\mathbb{Q}$ to C_Γ , the same residue-class argument reduces the unknown part of the functional kernel to linear relations among the Gamma monomials $\gamma^{\mathbf{a}}$. Thus the functional variable contributes only the Kummer-weight binomials above; the remaining arithmetic problem is the relation problem among the chosen Gamma constants, together with $2\pi i$ in the usual global formulation. In this sense, the theorem gives a fixed-denominator functional analogue of the Rohrlich–Lang picture while leaving its classical numerical core open.

10.3 Nilsson– E packets

A finite packet of functions

$$z^\alpha (\log z)^m F(z), \quad \alpha \in \mathbb{Q},$$

with F an E -function has finite Kummer and unipotent monodromy at the regular singularity. Monodromy projectors and finite differences provide a concrete split model for separating the Kummer and logarithmic components. Once the packet is placed in the relative exponential Nori envelope, the generic functional theorem applies. Relations at a fixed algebraic value with period-valued coefficients remain arithmetic and are not consequences of the functional theorem.

11 Concluding perspective

The relative exponential Nori category admits the same functional-period structure as the ordinary motivic setting. The fixed-part theorem identifies constant exponential submotives, while the Fourier nearby-cycle and 1-motivic arguments control differential characters. Together they give the relative Galois exact sequence and identify the generic comparison torsor with the Picard–Vessiot torsor.

For the filtered generic Kontsevich–Zagier exponential envelope, the formal functional period map is therefore injective at every algebraic nonzero Fourier scale. The cyclotomic Gamma calculation gives an explicit instance: over the field generated by the constant Gamma values, the complete functional ideal is the Kummer binomial ideal of Theorem 10.1, while the classical Rohrlich–Lang problem remains entirely in the constants. Arithmetic relations created by specialization and an elementary presentation of the general motivic relation algebra are separate problems.

AI-assisted research disclosure

The authors used AI-assisted tools for literature retrieval, drafting, and proof checking, and take responsibility for the manuscript. The AI system is not an author.

References

- [1] G. W. Anderson, W. D. Brownawell, and M. A. Papanikolas, *Determination of the algebraic relations among special Γ -values in positive characteristic*, Ann. of Math. (2) **160** (2004), no. 1, 237–313.
- [2] Y. Andre, *Mumford–Tate groups of mixed Hodge structures and the theorem of the fixed part*, Compositio Math. **82** (1992), no. 1, 1–24.
- [3] J. Ayoub, *Periods and the conjectures of Grothendieck and Kontsevich–Zagier*, Eur. Math. Soc. Newsl. (2014), no. 91, 12–18.
- [4] J. Ayoub, *Une version relative de la conjecture des periodes de Kontsevich–Zagier*, Ann. of Math. (2) **181** (2015), no. 3, 905–992.
- [5] J. Ayoub and L. Barbieri-Viale, *Nori 1-motives*, Math. Ann. **361** (2015), no. 1–2, 367–402.

- [6] L. Barbieri-Viale and V. Srinivas, *Albanese and Picard 1-motives*, Mém. Soc. Math. Fr. (N.S.) **87** (2001), vi+104 pp.
- [7] A. A. Beilinson, J. Bernstein, and P. Deligne, *Faisceaux pervers*, Asterisque **100** (1982).
- [8] B. Bakker and J. Tsimerman, *Functional transcendence of periods and the geometric Andre–Grothendieck period conjecture*, Forum Math. Sigma **13** (2025), e97.
- [9] J. Commelin, P. Habegger, and A. Huber, *Exponential periods and o-minimality*, arXiv:2007.08280, version 4, 2026.
- [10] J. Commelin and A. Huber, *Exponential periods and o-minimality II*, arXiv:2007.08290.
- [11] J. Fresan and P. Jossen, *Exponential motives*, preliminary manuscript, version consulted August 2026, available from the authors.
- [12] M. Gallauer and S. Pepin Lehalleur, *Exponentiation of coefficient systems and exponential motives*, arXiv:2211.17247.
- [13] A. Huber and G. Wüstholz, *Transcendence and Linear Relations of 1-Periods*, Cambridge Tracts in Mathematics, Cambridge University Press, 2022.
- [14] M. Hien, *Periods for flat algebraic connections*, Invent. Math. **178** (2009), no. 1, 1–22.
- [15] M. Hien and C. Roucairol, *Integral representations for solutions of exponential Gauss–Manin systems*, Bull. Soc. Math. France **136** (2008), no. 4, 505–532.
- [16] F. Ivorra and S. Morel, *The four operations on perverse motives*, J. Eur. Math. Soc. **26** (2024), 4191–4272.
- [17] P. Deligne, *Théorie de Hodge. III*, Publ. Math. Inst. Hautes Études Sci. **44** (1974), 5–77.
- [18] P. Deligne and J. S. Milne, *Tannakian categories*, in *Hodge Cycles, Motives, and Shimura Varieties*, Lecture Notes in Math. **900**, Springer, 1982, pp. 101–228.
- [19] A. Douai and C. Sabbah, *Gauss–Manin systems, Brieskorn lattices and Frobenius structures (I)*, Ann. Inst. Fourier (Grenoble) **53** (2003), no. 4, 1055–1116.
- [20] E. Jacobsen and L. Terenzi, *A comparison of categories of Nori motivic sheaves*, arXiv:2509.21476v1, 2025.
- [21] N. M. Katz, *Exponential Sums and Differential Equations*, Annals of Mathematics Studies **124**, Princeton University Press, 1990.
- [22] M. Kontsevich and D. Zagier, *Periods*, in *Mathematics Unlimited–2001 and Beyond*, Springer, 2001, pp. 771–808.
- [23] N. Koblitz and A. Ogus, *Algebraicity of some products of values of the Γ function*, appendix to P. Deligne, *Valeurs de fonctions L et périodes d’intégrales*, Proc. Sympos. Pure Math. **33**, part 2, Amer. Math. Soc., 1979, pp. 343–346.
- [24] Y.-H. J. Lam and D. Litt, *Geometric local systems on the projective line minus four points*, Compos. Math. **161** (2025), no. 3, 536–554.

- [25] A. Mostaed, *On the invariant cycle theorem for families of Nori motives*, arXiv:2206.05582.
- [26] R. Hartshorne, *On the de Rham cohomology of algebraic varieties*, Publ. Math. Inst. Hautes Etudes Sci. **45** (1975), 5–99.
- [27] R. Hotta, K. Takeuchi, and T. Tanisaki, *D-Modules, Perverse Sheaves, and Representation Theory*, Progress in Mathematics **236**, Birkhäuser, 2008.
- [28] T. Rivoal, *On the arithmetic nature of the values of the Gamma function, Euler’s constant, and Gompertz’s constant*, Michigan Math. J. **61** (2012), no. 2, 239–254.
- [29] M. Saito, *Mixed Hodge modules*, Publ. Res. Inst. Math. Sci. **26** (1990), no. 2, 221–333.
- [30] C. Sabbah, *Polarizable Twistor D-Modules*, Astérisque **300**, Société Mathématique de France, 2005.
- [31] C. Sabbah, *Monodromy at infinity and Fourier transform II*, Publ. Res. Inst. Math. Sci. **42** (2006), no. 3, 803–835.
- [32] C. Sabbah, *A short proof of a theorem of Cotti, Dubrovin and Guzzetti*, Port. Math. **79** (2022), no. 1, 29–43; arXiv:2103.16878v3.
- [33] B. Snodgrass, *Periods of E-operators*, arXiv:2608.06005, 2026.
- [34] L. Terenzi, *Tensor structure on perverse Nori motives*, Ann. K-Theory **11** (2026), no. 1, 47–170; arXiv:2401.13547v3.
- [35] S. Tubach, *On the Nori and Hodge realisations of Voevodsky motives*, Compos. Math. **161** (2025), no. 9, 2155–2201; arXiv:2309.11999v4.
- [36] R. Virk, *Realizations of exponential sheaves and Fourier transform*, arXiv:2605.14904v4, 2026.